

Supplemental Material for “A strict Rayleigh bound state in the continuum in a two-groove underwater acoustic metagrating”

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(Dated: August 24, 2026)

SCOPE, CONVENTIONS, AND REPRODUCIBILITY

The main Letter reports a theory result for an infinite, rigid, lossless periodic surface. This Supplemental Material gives the modal-matching operator, the strict Rayleigh criterion, numerical convergence, complex-pole tracking, field normalization, cavity ablation, dimensional scaling, fabrication perturbations, and driven-scattering controls. The final retained design is evaluated at $f_0 = 200$ kHz and $c_0 = 1500$ m/s. Pressure has time dependence $e^{+i\omega t}$; reflected propagating waves and evanescent waves use the outgoing/decaying square-root branch. Superscripts hom and dr distinguish homogeneous eigenmode amplitudes from driven scattering amplitudes.

The strict result uses $(N, K) = (313, 39)$, where N is the odd number of Floquet orders and K is the number of cosine modes per groove. The final same-geometry pole uses the same truncation. Separate lower- and higher-order calculations state their truncations explicitly. All numerical figures are generated from machine-readable MAT/CSV files. The figure drivers are listed in the data-and-code map below.

MODAL MATCHING

For a period a , incidence angle θ , and $k_0 = \omega/c$,

$$k_{x,n} = k_0 \sin \theta + \frac{2\pi n}{a}, \quad k_{y,n} = \sqrt{k_0^2 - k_{x,n}^2}, \quad (\text{S1})$$

where $\Re k_{y,n} \geq 0$ and $\Im k_{y,n} \leq 0$. With the surface at $y = 0$ and water in $y > 0$,

$$p_{\text{inc}} = e^{-ik_{x,0}x + ik_{y,0}y}, \quad p_{\text{sc}} = \sum_n A_n^{\text{dr}} e^{-ik_{x,n}x - ik_{y,n}y}. \quad (\text{S2})$$

The normalized variables are

$$\Omega = \frac{af}{c}, \quad \kappa = \Omega \sin \theta, \quad |\kappa + n| = \Omega \quad (\text{S3})$$

at the Rayleigh opening of order n .

Groove ℓ occupies $x_\ell < x < x_\ell + w_\ell$ and $-d_\ell < y < 0$. The bottom-rigid expansion is

$$p_\ell(x, y) = \sum_{q=0}^{K-1} C_{\ell q} \cos \left[\frac{q\pi(x - x_\ell)}{w_\ell} \right] \cos[\beta_{\ell q}(y + d_\ell)], \quad (\text{S4})$$

$$\beta_{\ell q}^2 = k_0^2 - \left(\frac{q\pi}{w_\ell} \right)^2. \quad (\text{S5})$$

The pressure projection used in the code is

$$\Pi_{n,\ell q} = \frac{1}{w_\ell} \int_{x_\ell}^{x_\ell + w_\ell} e^{-ik_{x,n}x} \cos \left[\frac{q\pi(x - x_\ell)}{w_\ell} \right] dx. \quad (\text{S6})$$

Cosine norms are 1 for $q = 0$ and $1/2$ for $q > 0$.

Let $\mathbf{B}$ map exterior pressure amplitudes to aperture cosine tests, and let $\mathbf{D}$ map groove normal velocity to global Floquet tests. With

$$\begin{aligned} \mathbf{C}_d &= \text{diag}[\cos(\beta_{\ell q} d_\ell)], \\ \mathbf{V}_d &= \text{diag}[(\beta_{\ell q}/k_0) \sin(\beta_{\ell q} d_\ell)]. \end{aligned} \quad (\text{S7})$$

the pole-free homogeneous operator is

$$\mathbf{F} \begin{bmatrix} \mathbf{A}^{\text{hom}} \\ \mathbf{C} \end{bmatrix} = 0, \quad \mathbf{F} = \begin{bmatrix} \mathbf{B} & -\mathbf{C}_d \\ \text{diag}(k_{y,n}/k_0) & i\mathbf{D}\mathbf{V}_d \end{bmatrix}. \quad (\text{S8})$$

The $C_{\ell q}$ are retained as unknowns. This avoids the tangent poles introduced if the groove variables are eliminated before forming the homogeneous problem.

For the forced problem the eliminated form is convenient. Define

$$\mathbf{S} = \mathbf{D} \text{diag}[-i\beta_{\ell q} \tan(\beta_{\ell q} d_\ell)]. \quad (\text{S9})$$

The reflected amplitudes obey

$$[\text{diag}(k_{y,n}) - \mathbf{S}\mathbf{B}] \mathbf{A}^{\text{dr}} = k_{y,0} \mathbf{e}_0 + \mathbf{S}\mathbf{b}_{\text{inc}}. \quad (\text{S10})$$

For each propagating order,

$$\eta_n = \frac{\Re k_{y,n}}{k_{y,0}} |A_n^{\text{dr}}|^2, \quad \sum_{n \in \text{prop}} \eta_n = 1 \quad (\text{S11})$$

up to numerical error in the lossless model.

Row/column scaling and reported residuals

Let $\mathbf{F}_{\text{red}}$ be the operator after deleting the selected threshold amplitude and every finite-flux open amplitude. Row and column scales are the corresponding Euclidean norms with a finite numerical floor,

$$\tilde{\mathbf{F}} = \mathbf{R}^{-1} \mathbf{F}_{\text{red}} \mathbf{C}^{-1}. \quad (\text{S12})$$

If $\mathbf{y}$ is the last right singular vector of $\tilde{\mathbf{F}}$, the physical reduced vector is $\mathbf{u} = \mathbf{C}^{-1}\mathbf{y}$. We report

$$\sigma_{\text{rel}} = \frac{\sigma_{\text{min}}}{\sigma_{\text{max}}}, \quad r_{\text{raw}} = \frac{\|\mathbf{F}_{\text{red}}\mathbf{u}\|_2}{\|\mathbf{u}\|_2}. \quad (\text{S13})$$

These are numerical compatibility diagnostics in scaled and unscaled coordinates, not two independent observables.

STRICT RAYLEIGH CONDITION AND ENDPOINT AUDIT

For a homogeneous threshold order, $k_{y,n} = 0$ and

$$\int_0^H |p_n^{\text{hom}}|^2 dy = H |A_n^{\text{hom}}|^2. \quad (\text{S14})$$

For an evanescent order $k_{y,n} = -i\alpha_n$,

$$\int_0^\infty |p_n^{\text{hom}}|^2 dy = \frac{|A_n^{\text{hom}}|^2}{2\alpha_n}. \quad (\text{S15})$$

At positive angle the strict constraint is therefore

$$A_{-1}^{\text{hom}} = A_0^{\text{hom}} = 0. \quad (\text{S16})$$

The final normalized geometry is

$$\begin{aligned} \kappa &= 0.083848696553, & g/a &= 0.134862478208, \\ d_1/a &= 0.660941508245, & d_2/a &= 0.157263731977, \\ w_1/a &= 0.577018249316, & w_2/a &= 0.153256794269. \end{aligned} \quad (\text{S17})$$

At $\Omega = 1 - \kappa$, the (313, 39) strict operator gives $\sigma_{\text{rel}} = 1.2603643 \times 10^{-15}$ and $r_{\text{raw}} = 1.3765668 \times 10^{-13}$.

The target column is always explicitly identified by order, but floating evaluation of the physical square root gives $k_{y,-1} = 1.1921 \times 10^{-7}$ in the normalized final calculation, whereas the original automatic threshold tolerance was 5.7563×10^{-8} . This can label the target as a tiny positive open channel even though $\Omega = 1 - \kappa$ exactly. All final endpoint audits therefore use the exact normalized Rayleigh relation and either an explicit threshold tolerance of 10^{-6} in normalized inverse-length units or the complex operator with target $k_{y,-1}$ inserted as exactly zero. The latter full, unconstrained calculation gives

$$\sigma_{\text{rel}}^{\text{full}} = 7.7151138 \times 10^{-16}. \quad (\text{S18})$$

Applying the full operator to the reconstructed strict vector gives the raw residual $1.3765668 \times 10^{-13}$, consistent with Eq. (S13).

REOPTIMIZED ROOTS AND FIXED-GEOMETRY CROSSCHECK

Table S1 gives the continued sequence of independently reoptimized roots. At each point the threshold and finite-open errors are extracted from the full, unconstrained

null vector and normalized by the groove coefficient norm. The root parameters approach a limiting sequence; the largest parameter change from one listed root to the next is 7.765×10^{-6} at the final step.

The same highest- K geometry is also evaluated without reoptimization at $K = 35, 39, 43, 47, 51$. Its strict singular-value ratios are 6.53×10^{-5} , 3.83×10^{-5} , 1.71×10^{-5} , 6.25×10^{-16} , and 1.40×10^{-5} , respectively. This fixed crosscheck distinguishes a root sequence from the stronger, false statement that one discretized geometry is an exact null at every truncation.

FINAL-GEOMETRY COMPLEX-POLE CONTINUATION

The superseded pole cache used an intermediate (121, 15) root and is not used in the revised figures. The reported continuation starts from the final (313, 39) geometry. For target order $n = -1$, a local target normal wavenumber $\bar{q}$ defines

$$\Omega_p = \sqrt{(\kappa - 1)^2 + \bar{q}^2}, \quad Q = \frac{\Re \Omega_p}{2|\Im \Omega_p|}. \quad (\text{S19})$$

Thirteen nonzero points span $10^{-5} \leq |\Delta\kappa| \leq 10^{-3}$, followed by the exact endpoint. All accepted complex roots have singular-value ratio below 10^{-7} ; the minimum consecutive mode overlap is 0.999999500, and no branch jump is detected. Over $3 \times 10^{-5} \leq |\Delta\kappa| \leq 10^{-3}$,

$$\log Q = (-1.99590867) \log |\Delta\kappa| + \text{const.} \quad (\text{S20})$$

The nearest finite value is $Q = 2.02613 \times 10^{12}$ at $|\Delta\kappa| = 10^{-5}$. Equation (S20) is a local finite-model fit, not a universal Rayleigh-BIC exponent.

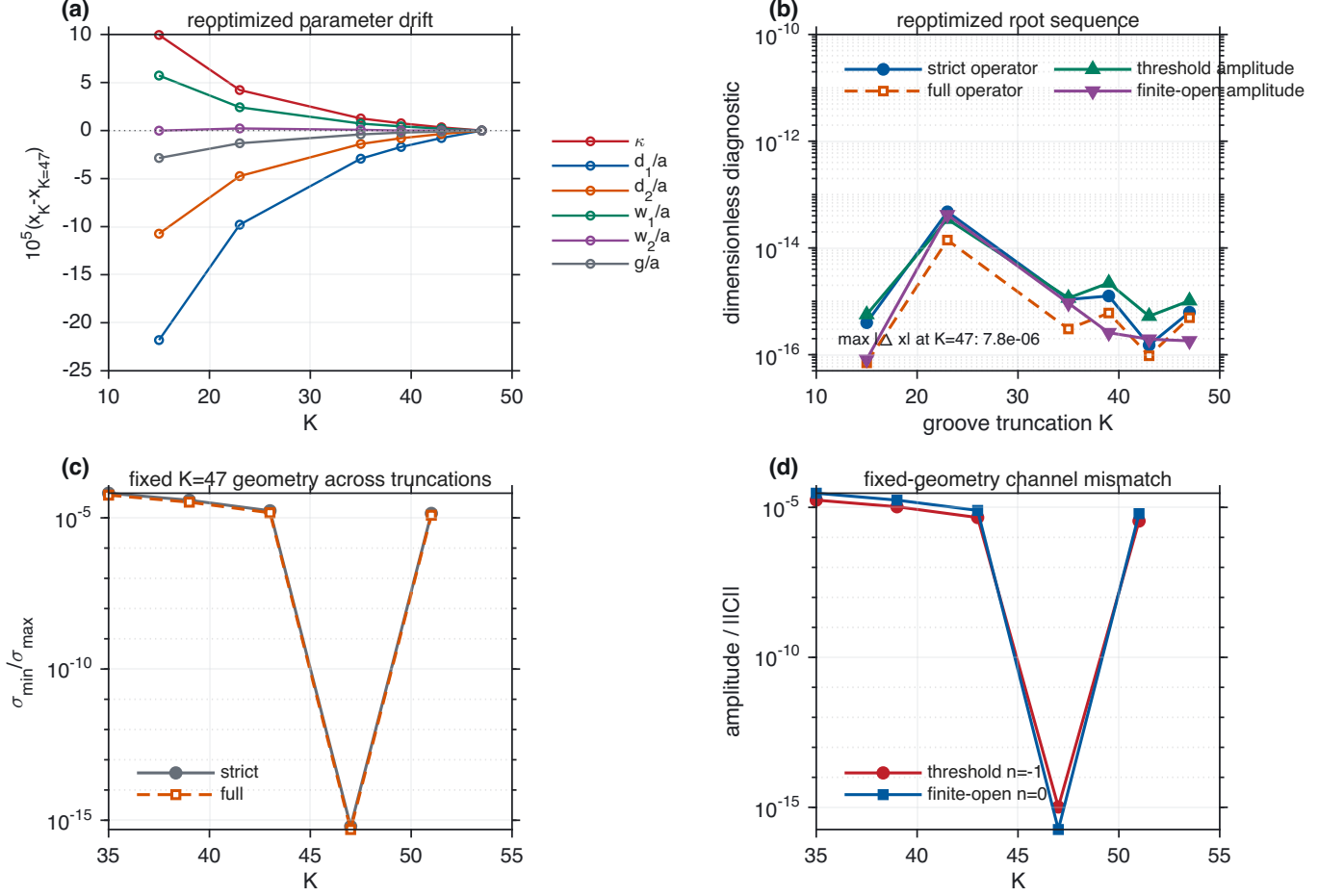
HOMOGENEOUS-FIELD CONSTRUCTION

All three main-text field maps use the same grid $0 \leq x/a \leq 1$, $-0.75 \leq y/a \leq 1.05$ and the same color range. Each coefficient vector is normalized in Euclidean norm; the plotted pressure is then divided by its maximum magnitude inside the grooves. The strict BIC coefficients are obtained at (313, 39) and truncated to (121, 15) only for stable field sampling. The other two fields are last right singular vectors of the complete real-frequency homogeneous operator and are explicitly surrogates rather than exact quasinormal modes.

For the strict state the exterior norm proxy reaches 0.017690 by $H = a$ and 0.017702 by $H = 5a$, remaining unchanged through $20a$. Adding a nonzero grazing component produces the expected linear term and increases the same proxy to 0.067702 at $20a$.

TABLE S1. Reoptimized strict-root sequence. A dash denotes the first point, for which no preceding parameter step exists.

(N, K)	κ	σ_{strict}	σ_{full}	$ A_{-1} / C $	$ A_0 / C $	$\max \Delta x $
(121, 15)	0.083940637	3.98×10^{-16}	7.03×10^{-17}	5.65×10^{-16}	8.20×10^{-17}	—
(185, 23)	0.083883325	4.76×10^{-14}	1.41×10^{-14}	3.47×10^{-14}	4.19×10^{-14}	1.20×10^{-4}
(281, 35)	0.083853785	1.09×10^{-15}	3.03×10^{-16}	1.15×10^{-15}	9.25×10^{-16}	6.87×10^{-5}
(313, 39)	0.083848697	1.26×10^{-15}	6.04×10^{-16}	2.21×10^{-15}	2.59×10^{-16}	1.22×10^{-5}
(345, 43)	0.083844523	1.51×10^{-16}	9.55×10^{-17}	5.27×10^{-16}	1.95×10^{-16}	9.30×10^{-6}
(377, 47)	0.083841177	6.25×10^{-16}	4.94×10^{-16}	1.03×10^{-15}	1.82×10^{-16}	7.77×10^{-6}

FIG. S1. Detailed modal convergence. (a) Reoptimized parameter drift relative to the (377, 47) root. (b) Strict/full residuals and unconstrained radiation amplitudes at the reoptimized roots. (c,d) Fixed highest- K geometry evaluated at neighboring truncations. The latter panels are a discretization crosscheck, not a failure of the independently reoptimized sequence.

SINGLE-CAVITY ABLATION AND TWO RADIATION PHASORS

The single-cavity comparison is a declared numerical search, not a theorem. It varies $x = (\kappa, d/a, w/a)$ over

$$\begin{aligned}
 0.04 &\leq \kappa \leq 0.25, \\
 0.04 &\leq d/a \leq 2.00, \\
 0.50 &\leq w/a \leq 0.95.
 \end{aligned} \tag{S21}$$

at the exact $n = -1$ Rayleigh relation $\Omega = 1 - \kappa$. A deterministic 1800-point survey and 14 bounded local refinements are followed through $(N, K) = (221, 44)$. The best final point is $(\kappa, d/a, w/a) = (0.04, 0.520817128, 0.530693214)$, with $\sigma_{\text{rel}} = 0.1245882$, raw residual 1.48399, and transverse coefficient fraction 0.961. Extra transverse modes therefore do not close both channels in this large rectangular-groove family. The lower aperture bound $w/a = 0.50$ is part of the ablation definition; arbitrary narrow, compound, or internally

TABLE S2. Homogeneous-field diagnostics. “Exterior/groove” is a finite-window normalized energy proxy, not an absolute modal energy.

state	Ω	θ (deg)	σ_{rel}	$ A_{-1} /\ A\ $	$ A_0 /\ A\ $	exterior/groove
strict BIC	0.916151303	5.25122	1.26×10^{-15}	0	0	0.0742
near-BIC surrogate	0.915551303	5.25467	2.53×10^{-3}	0.0203	0.00262	0.0736
ordinary surrogate	0.865500000	5.55947	1.31×10^{-3}	0.886	0.0697	0.752

structured one-cavity designs are outside its scope.

For the two-groove mode, each radiation row can be decomposed as

$$A_m^{\text{hom}} = \sum_{\ell,q} \mathcal{R}_{m,\ell q} C_{\ell q}, \quad m = -1, 0. \quad (\text{S22})$$

The unnormalized two-groove sums are 1.60×10^{-16} for $n = -1$ and 1.06×10^{-16} for $n = 0$; after normalization by the largest individual groove contribution, their closures are 1.6×10^{-15} and 1.1×10^{-15} . This supports the two-phase interpretation within the searched family.

DIMENSIONAL SCALING AND SOUND-SPEED CONTROL

With all normalized dimensions and θ fixed, set $a = sa_0$. The exact similarity law is

$$f_{\text{BIC}}(c, s) = \frac{\Omega_0 c}{sa_0} = f_0 \frac{c}{c_0} \frac{1}{s}. \quad (\text{S23})$$

The modal operator is rebuilt at every dimensional point. For $c = 1400 : 10 : 1600$ m/s with $s = 1$, f_{BIC} spans 186.666667–213.333333 kHz and the strict raw residual remains between 9.69×10^{-14} and 1.14×10^{-13} . For $s = 0.5 : 0.1 : 2$ with $c = c_0$, the frequency spans 400–100 kHz and the raw residual remains between 9.16×10^{-14} and 1.86×10^{-13} .

As a control, the geometry, $f = 200$ kHz, and θ are then held fixed while c changes. Only $c = 1500$ m/s remains at the Rayleigh threshold. The full-operator raw residual reaches 0.649 and the open-amplitude ratio reaches 0.776 over the sweep. The retuned similarity result must therefore not be called temperature or environmental robustness of a fixed sample.

FABRICATION PERTURBATIONS

The fabrication ensemble keeps the final period, frequency, sound speed, and Bloch point fixed and perturbs the five normalized dimensions. For each bound $p = 1\%, 2\%, 5\%$:

- the “one-parameter” ensemble contains 100 balanced samples, cycling through d_1, d_2, w_1, w_2, g and drawing the selected error uniformly from $[-p, p]$;

- the “joint” ensemble contains 100 samples with all five errors drawn independently and uniformly from $[-p, p]$.

The random seed is 260824 and the ensemble truncation is (185, 23). All 600 geometries are valid and all solves finish. The unperturbed final geometry at this truncation has strict singular-value ratio 1.8210×10^{-4} and raw residual 2.2198×10^{-3} , which is recorded as the discretization floor.

The strict raw residual is evaluated after enforcing $A_{-1}^{\text{hom}} = A_0^{\text{hom}} = 0$. In contrast, the threshold error reported below is the physically informative unconstrained value

$$\epsilon_{-1} = \frac{|A_{-1}|}{\|A\|_2}, \quad \epsilon_0 = \frac{|A_0|}{\|A\|_2}. \quad (\text{S24})$$

The strict operator’s target amplitude is exactly zero by construction and is not used as a tolerance metric.

DRIVEN SCATTERING: RECIPROCAL BUT NOT CONVERGED ROUTING

Driven scattering is not used to certify the homogeneous BIC. It is included to audit potential observables and to expose the truncation sensitivity of the previous route point. The final geometry is held fixed at $f = 200216.874212$ Hz and $\theta = +6^\circ$. Direct solver results are listed in Table S4; no cached efficiency table is used.

The efficiency is plainly nonconverged at fixed geometry and frequency. Energy closure remains below 10^{-13} , so the variation is not hidden power loss; it is a truncation/conditioning problem. At (313, 39), reversal to -6° exchanges the signed target order:

$$\begin{aligned} \eta_{-1}(+6^\circ) &= 0.306061126315, \\ \eta_{+1}(-6^\circ) &= 0.306061126315. \end{aligned} \quad (\text{S25})$$

with difference 1.10×10^{-13} . The output angles are $\mp 80.336964^\circ$. This is reciprocal signed-order exchange, not an isolator or a converged one-way channel.

A separate ordinary point at $f = 202430$ Hz and $\theta = 32.6328^\circ$ gives target efficiencies 0.9779112, 0.9929755, 0.9986358, 0.9996289, 0.9999151, and 0.9996185 for (81, 11) through (401, 50). The final nonmonotonic step

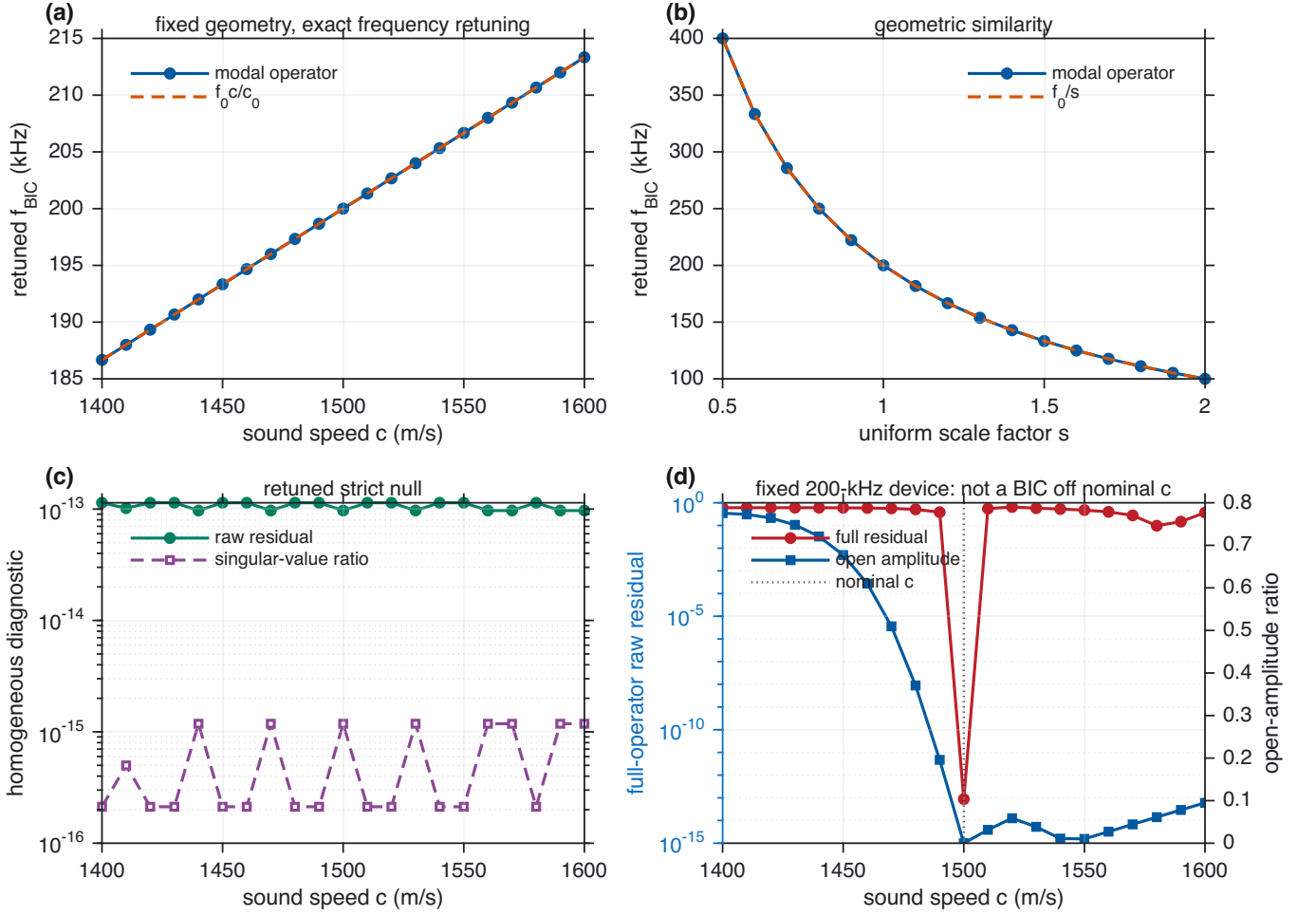


FIG. S2. Dimensional similarity and fixed-device control. (a) BIC frequency versus sound speed under exact frequency retuning. (b) Inverse scaling with uniform geometric factor. (c) Retuned homogeneous residuals remain at numerical precision. (d) At fixed geometry, frequency, and angle, changing sound speed detunes the full operator and reopens radiation; off nominal c these points are not Rayleigh BICs.

TABLE S3. Five-parameter tolerance statistics. Entries are 5th/50th/95th percentiles.

ensemble	bound	strict raw residual	unconstrained ϵ_{-1}
one parameter	$\pm 1\%$	0.00250 / 0.0136 / 0.429	0.000140 / 0.00315 / 0.214
one parameter	$\pm 2\%$	0.00440 / 0.0359 / 0.611	0.000930 / 0.00784 / 0.432
one parameter	$\pm 5\%$	0.0151 / 0.0696 / 0.591	0.00411 / 0.0194 / 0.809
joint five parameter	$\pm 1\%$	0.0595 / 0.351 / 0.526	0.00951 / 0.162 / 0.273
joint five parameter	$\pm 2\%$	0.183 / 0.580 / 0.842	0.0604 / 0.338 / 0.522
joint five parameter	$\pm 5\%$	0.302 / 0.576 / 1.44	0.0866 / 0.637 / 0.948

is retained. At $(401, 50)$, $\eta_{-1} = 0.9996185039$, $\eta_0 = 0.0003814961$, condition number 7.32×10^3 , and energy error 8.9×10^{-15} . Its active Rayleigh frequency is approximately 141.825 kHz; the state is therefore off Rayleigh and not BIC-enhanced.

PROPOSED FUTURE WATER-TANK VALIDATION

No experiment is reported in the Letter. Figure S5 records a falsifiable protocol for a later finite-sample study. The proposed sample set includes the full two-groove array, a flat rigid reference, a single-large-cavity control, and the two-groove cell with the small cavity

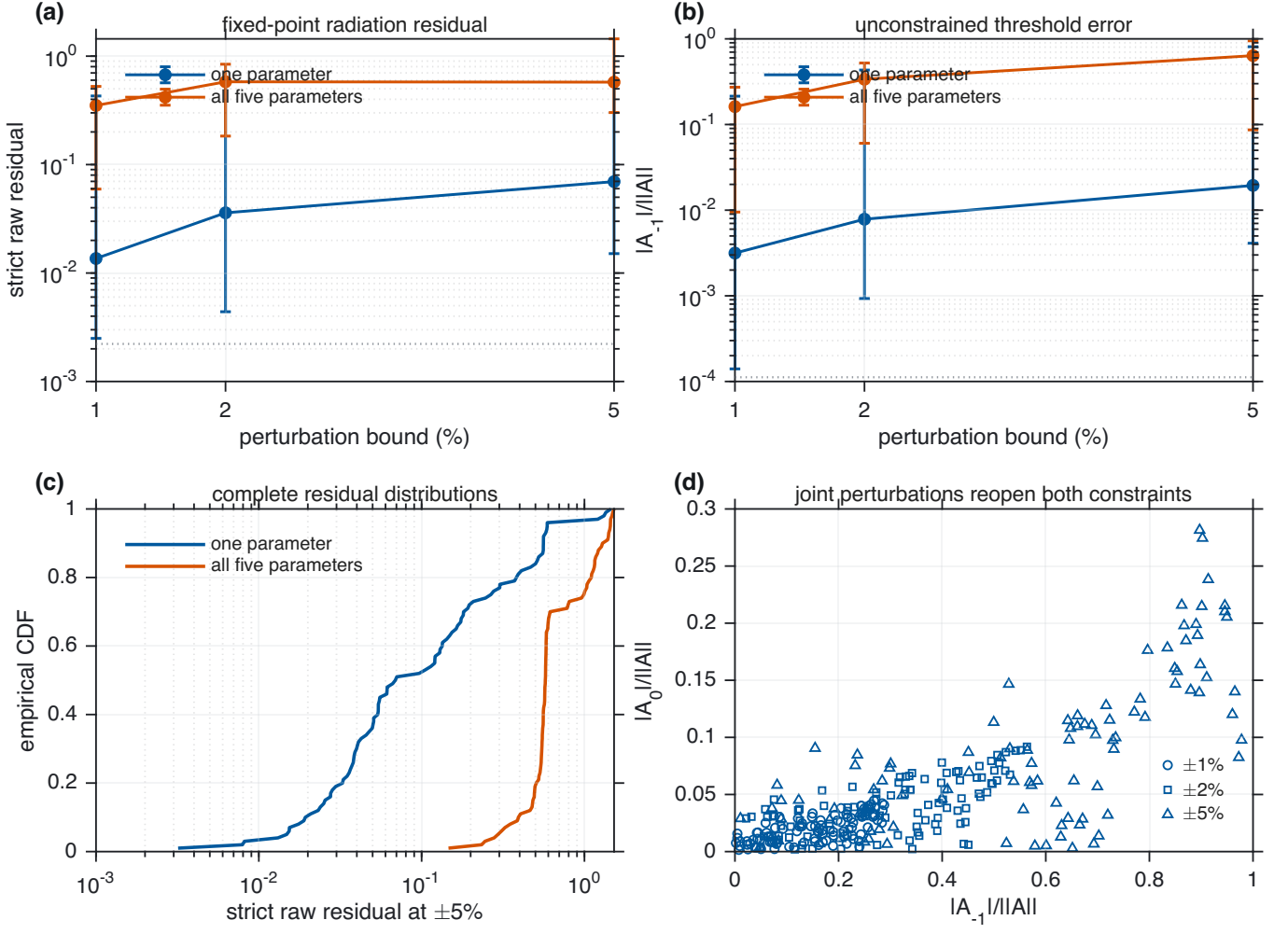


FIG. S3. Fabrication statistics at fixed Rayleigh point. (a,b) Residual and unconstrained threshold error; markers are medians and bars are 5–95% intervals. (c) Full $\pm 5\%$ residual distributions. (d) Joint perturbations reopen both the threshold and finite-flux amplitudes. These uniform local ensembles are sensitivity diagnostics, not a manufacturing process model or yield prediction.

TABLE S4. Fixed-geometry driven route. The target is $n = -1$ for positive incidence.

(N, K)	η_{-1}	η_0	condition number	energy error
(81, 11)	0.00009184	0.9999082	1.95×10^4	1.1×10^{-16}
(121, 15)	0.00040379	0.9995962	4.68×10^4	1.0×10^{-15}
(201, 25)	0.00659293	0.9934071	2.61×10^5	1.3×10^{-15}
(313, 39)	0.30606113	0.69393887	2.59×10^6	3.7×10^{-14}
(401, 50)	0.05715278	0.94284722	1.41×10^6	3.2×10^{-14}

blocked. Several array lengths should be compared before interpreting a linewidth as an infinite-period Q .

The water sound speed should be measured by time of flight. The source angular spectrum, complex hydrophone transfer function, and flat-plate reflection should be calibrated. Complex pressure can be sampled

over several periods at height y_0 and fitted as

$$p_{\text{sc}}(x, y_0) = \sum_n \tilde{A}_n(y_0) e^{-ik_x n x},$$

$$A_n = \tilde{A}_n(y_0) e^{+ik_y n y_0}. \quad (\text{S26})$$

The fitted amplitudes are then converted to normal-power efficiencies with Eq. (S11). A grazing pressure feature must be reported separately from its zero normal

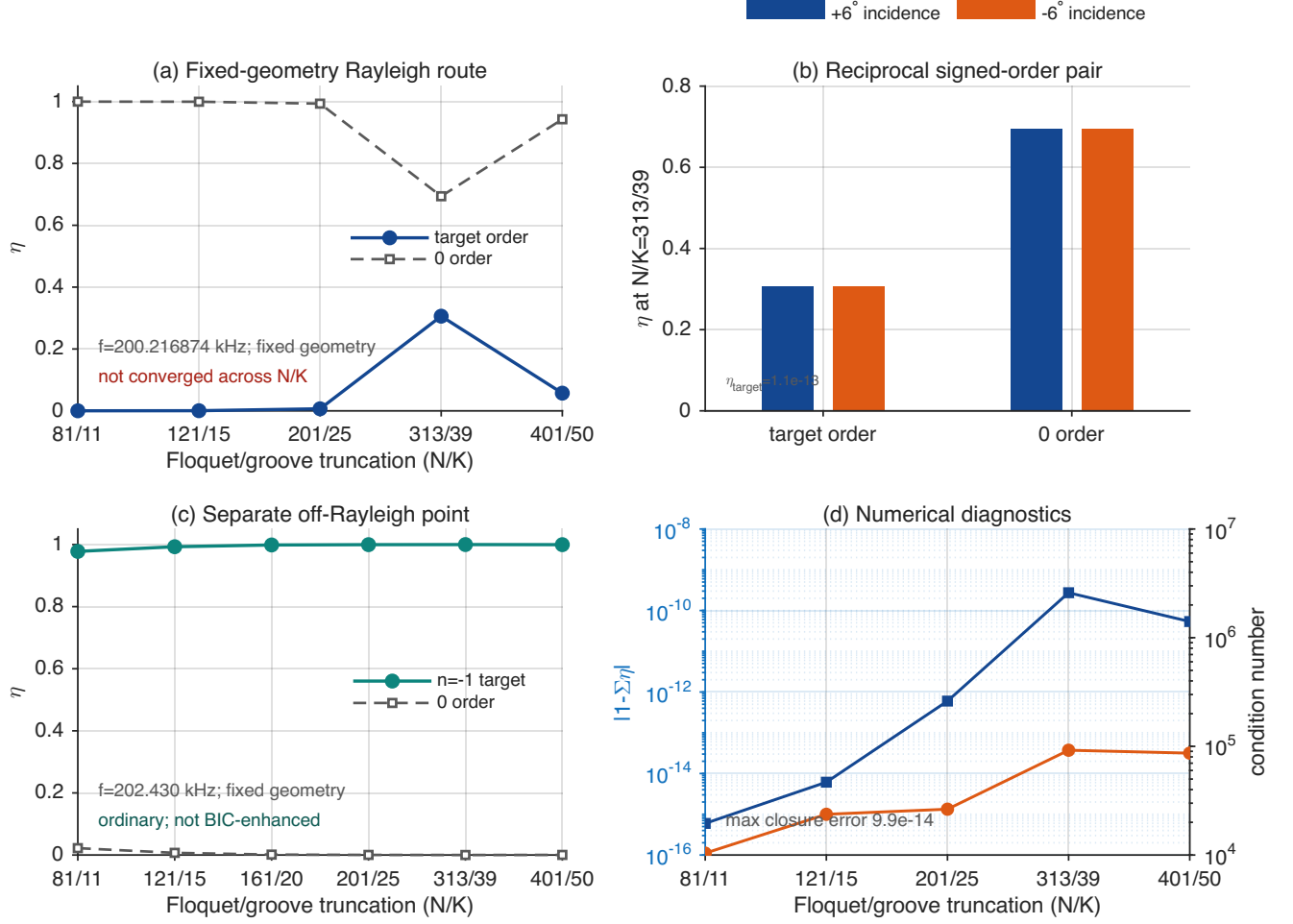


FIG. S4. Driven-scattering audit. (a) Fixed-geometry route efficiency versus modal truncation. (b) Reciprocal signed-order pair at the illustrative (313, 39) point. (c) Separate off-Rayleigh ordinary reflector. (d) Energy-closure error and matrix condition number for the route sweep. Panel (a) precludes a claim of converged device efficiency.

power.

For a narrowband burst, an amplitude ring-down $p(t) \propto \exp(-t/\tau_A)$ would give

$$Q_{\text{ring}} = \pi f \tau_A. \quad (\text{S27})$$

Finite width, source aperture, viscothermal attenuation, elastic backing, and dimensional uncertainty must enter the comparison. A driven reflection minimum alone cannot validate Eq. (S16).

DATA AND CODE MAP

The principal reproducible entry points are:

- `advanced_analysis/final_root_pole_validation.m`;

- `advanced_analysis/extended_convergence.m`;
- `advanced_analysis/scaling_environment.m`;
- `advanced_analysis/robustness_fields.m`;
- `advanced_analysis/driven_scattering_audit.m`;
- `advanced_analysis/run_prl_completion_figures.m`.

Machine-readable CSV/MAT outputs are stored in the corresponding `advanced_analysis/` subdirectories. The final design cache is `Ni2019_MATLAB/results/StrictRayleighBIC_200kHz_min1mm.mat`. The superseded intermediate pole cache and the previous hard-coded router convergence table are not used in this revised evidence set.

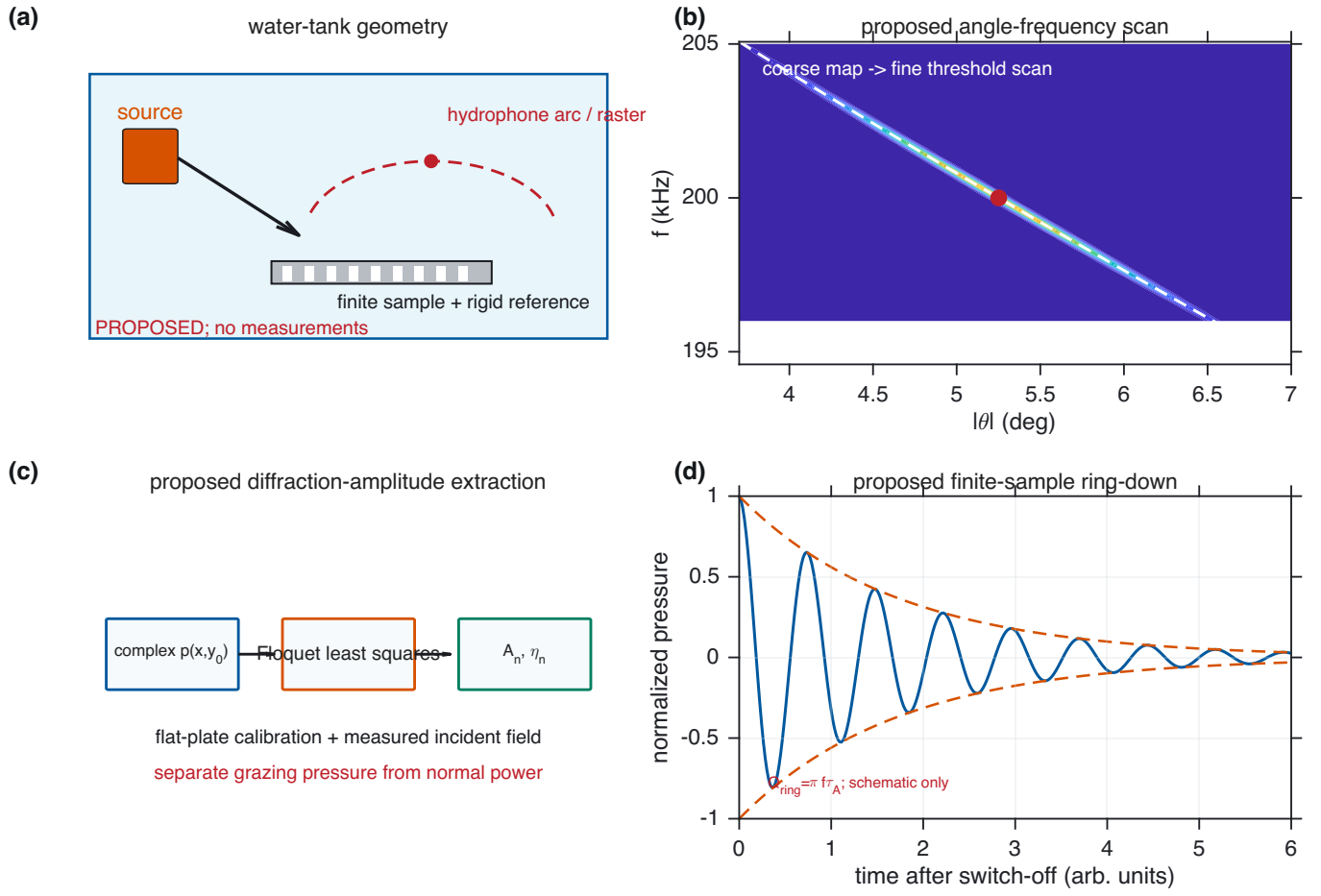


FIG. S5. PROPOSED future protocol; no measured data are shown. (a) Water-tank geometry with finite sample, flat reference, source, and hydrophone scan. (b) Coarse and fine angle-frequency scan around the threshold. (c) Complex-field Floquet extraction. (d) Finite-sample ring-down schematic.